Project Title:

ParkNow-Real Time Parking Slot Booking System

Batch No: 7

Abstract

Urban areas often face serious parking challenges due to the increasing number of vehicles and limited parking spaces. Drivers usually spend a significant amount of time searching for available parking, which leads to traffic congestion, fuel wastage, and frustration. Traditional parking systems do not provide real-time information about available parking slots or allow advance reservations.

The **ParkNow – Real-Time Parking Slot Booking System** is a web-based application designed to provide a smart solution for urban parking management. The system enables users to view available parking slots in real time and reserve them in advance through an online platform. It also provides administrators with tools to manage parking areas, monitor slot availability, and track reservations efficiently.

Objectives:

1. To provide real-time information about available parking slots.
2. To allow users to reserve parking spaces in advance through an online platform.
3. To reduce traffic congestion caused by vehicles searching for parking.
4. To maintain secure user authentication and booking records.
5. To improve convenience and time efficiency for drivers in urban areas.

Modules:

Module 1: User Authentication Module

Purpose: The User Authentication Module manages the registration and login process for users. It ensures that only authorized users can access the parking reservation system.

Features:

- User registration
- Secure login and logout
- Password protection
- Authentication verification

Module 2: Parking Area Management Module

Purpose: This module allows the administrator to manage parking locations and maintain the details of available parking spaces.
Features:

- Add new parking areas
- Update parking location information
- Define total number of parking slots
- Delete or modify parking areas

Module 3: Parking Slot Availability Module

Purpose:
The Parking Slot Availability Module displays the current status of parking spaces and helps users identify available slots easily.
Features:

- Display available parking slots
- Show reserved or occupied slots
- Real-time slot status updates
- Visual representation of parking availability

Module 4: Parking Reservation Module

Purpose:
This module allows users to select and reserve parking slots for a specific time and location.
Features:

- Select parking location
- Choose available slot
- Reserve parking space
- Store booking information in the database
- Confirm reservation to the user

Module 5: Booking History Module

Purpose:
The Booking History Module maintains a record of all reservations made by users.
Features:

- View previous bookings
- Display booking details
- Show reservation status
- Cancel or manage bookings

Module 6: Admin Dashboard Module

Purpose: The Admin Dashboard provides the administrator with complete control over the parking system and helps monitor overall activity.

Features:

- View total parking slots
- Monitor reservations
- Manage parking areas
- Track booking statistics
- Update parking slot status

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